Andrew Taggart

Data Scientist · Machine Learning · Analytics

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Summary

Data Science M.S. graduate (GPA 3.87) who turns raw data into models, dashboards, and decision tools. Builds end-to-end pipelines in Python, SQL, and Spark, with hands-on work in machine learning, NLP, time-series forecasting, and deep learning. Brings graduate research experience, a software engineering background, and earlier work as a supply-chain analyst turning forecasts and supplier data into monthly reports for procurement and engineering leaders. Communicates findings clearly to both technical and business audiences.

Technical Skills

Languages: Python, SQL, R, JavaScript | **Machine learning:** scikit-learn, PyTorch, TensorFlow/Keras, Spark MLlib, SHAP, NLTK, gensim; classification, forecasting (ARIMA), NLP, federated learning, model evaluation | **Data and big data:** pandas, NumPy, PySpark, Databricks, statsmodels, ETL, data cleaning, exploratory data analysis | **Databases:** PostgreSQL, SQLite, MongoDB| **Visualization:** Tableau, Matplotlib, Seaborn, Plotly/Dash, Streamlit, folium, Leaflet, Excel | **Engineering:** Git/GitHub, Docker, Jupyter, REST APIs, React, Django, Node.js

Research & Projects

Federated Healthcare Explainability Research | PyTorch, pandas, scikit-learn, SHAP

- Built FedAvg federated-learning experiments in PyTorch across 2-client and 5-client settings for classification and survival prediction.
- Wrote reusable preprocessing and evaluation pipelines with stratified splits, encoding, standardization, Macro-F1, C-index, and early stopping.
- Compared feature-attribution stability across centralized, global, and local models with permutation importance and SHAP.

NJ-EQUIP: Energy Equity Analytics Portal | React, Leaflet, data visualization | Best Paper, IEEE CCWC 2026

- Led development of an interactive portal that turns environmental, utility, and demographic data into maps and visual tools.
- Built an energy-burden estimator from utility costs, heating costs, and household income; turned researchers' requirements into features.

Tariff News Sentiment Analysis (team of 5) | Python, NLTK, gensim, scikit-learn, SQLite, Streamlit

- Built an NLP pipeline that pulls news articles from NewsAPI into SQLite and scores sentiment with SentiWordNet.
- Trained TF-IDF + Naive Bayes classifiers, modeled topics with LDA, and served predictions in a Streamlit app; found 61% of 96 articles positive.

Income Classification with Spark MLlib | PySpark, Spark MLlib

- Built a distributed ML pipeline on 46K census records with one-hot encoding and 5-fold cross-validated grid search.
- Reached a test ROC AUC of 0.907 with gradient-boosted trees, beating logistic regression (0.899).

US Retail Sales Forecasting | statsmodels, pmdarima

- Forecast 30+ years of monthly retail sales with ARIMA models chosen from ADF, ACF, and PACF diagnostics (MAPE 0.73%); added a seasonal Auto-ARIMA with economic indicators as inputs.

Experience

Research Assistant, Montclair State University (Prof. Chao Huang) | Montclair, NJ | Sep 2025 – May 2026

- Supported research on mitigating catastrophic forgetting in split learning and federated learning.
- Co-authored "Tackling Catastrophic Forgetting in Split Learning via Orthogonal Projection" (ICANN 2026).

Global Electronics Sourcing Analyst, Milwaukee Tool (TTI) | Brookfield, WI | Jul 2021 – Jul 2022

- Compared monthly semiconductor requirements with supplier commitments to flag shortages months ahead and guide procurement and engineering decisions.
- Built monthly stakeholder reports combining demand, committed supply, deficits, and alternative coverage to support allocation and delivery planning.
- Analyzed cost and usage forecasts and worked with engineers and suppliers to confirm lead times and approved alternatives during the global chip shortage.

Field Sales and Marketing Representative, Techtronic Industries (TTI) | Delray Beach, FL | Jul 2020 – Jun 2021

- Analyzed monthly sales and market potential to find growth opportunities and guide merchandising across partner stores.

Additional experience: Coach, Body Fit Training (2024–2025); Personal Trainer, Equinox (2022–2023). Built client relationships, tracked progress, and supported sales.

Education

- **M.S., Data Science**, Montclair State University | GPA 3.87 | May 2026
- **Software Engineering Immersive**, General Assembly | Apr 2024
- **B.S., Human and Social Development**, University of Miami | Minor in Exercise Physiology | May 2020

Publications & Awards

- Co-author, "Tackling Catastrophic Forgetting in Split Learning via Orthogonal Projection," ICANN 2026
- Co-author, Best Paper Award, IEEE CCWC 2026, for the NJ-EQUIP energy-equity project